Sparsh Sinha

Phone: +919205759876
Email: sparshsinha.spavis@gmail.com LinkedIn | Github

Education

B.Tech. in COE	2023	Delhi Technological University (DTU)	8.83 CGPA
CBSE (Class XII)	2018	Abhinav Public School, Delhi	90.4%

Work Experience

SDE II - Pocketpills *Jul 2023 - Current*

- Built the entire backend of a **multi-tenant** EMR **SaaS application**, featuring personalized dashboards and an AI-driven prescribing utility, for online and in-person consultation, using **Spring Boot, PostgreSQL, RabbitMQ,** and **MCP.**
- Built a **Questionnaire engine** and **creation tool** in Spring using **Graph and Binary Tree** algorithms, automating clinic workflows by replacing manual doctor-driven processes. Supports **10+ campaigns with 50+ dynamic questions**, collecting medical histories, and suggesting medications
- Implemented a real-time delivery module with **RabbitMQ, Camunda, and Spring**, cutting profile completion TAT by 40%. Utilized **WebClient and CompletableFutures** to optimize response times for high-volume, simultaneous requests, achieving an **improvement of 80%** in response time.
- Developed a personalized, interactive **backend-driven dashboard** with **100+ dynamic widgets**, attracting over 1 million monthly visits and driving a 25% boost in user engagement while **reducing inbound communications by 35%.**
- Created a customizable campaign scheduling tool for personalized messaging via FCM, Twilio, and SendGrid, supporting over **1,000 targeted campaigns sent to a 12M+ user base** across multiple channels (**email, SMS, push notifications**).
- Developed a Spring Boot-powered portal for B2B partners using **AWS S3** to upload and process census files, facilitating data collection for **100+ companies** and supporting a user base of over 2 million.

Backend Development Intern - Xtreme Technologies *Apr 2022 - Jun 2022*

- Created the backend of Xtreme's official website and the user dashboard using **Django, JWTs,** and **PostgreSQL**, averaging over **50k+ monthly visits.**
- **Django REST Framework** was used to design APIs to collect real-time and forecasted weather data at scheduled intervals
- Designed and implemented an **algorithm** to decrypt grib weather files and utilize them for **real-time voyage optimization** with an **accuracy of over 94%** and encrypt the output csv into binary krup files using **Python** and **C++**

Academic Projects

Intra-Organisational Server | (Github link) | *June 2022*

- Developed a highly secure, private, and scalable communication channel using **socket programming** in Dart.
- **Inter-device communication** is established by connecting all devices to a main server post-user authentication.
- Designed the front end of the mobile app using **Flutter.**

Chat App - MERN Stack | (Github link) | (Website Link) | *Mar 2022*

- Created a full-stack chat app using **Node.js, Express, and Socket.io**
- **Mustache** was used for templating the front end and the website was hosted on Heroku.
- Designed a highly scalable architecture facilitating vertical and horizontal scaling to accommodate a large user base.

Achievements

- **CodeKaze by Coding Ninjas -** Secured a rank of 315 among 2 lakh+ participants *Mar 2022*
- **Hack with CW Hackathon -** Finished in the top 30 among 800 teams *Dec 2020*

Positions of Responsibility

Joint Secretary – Sahitya DTU *Aug 2019 – Jun 2022*

- Responsible for doubling the social media presence of society from 600 to 1200+
- Responsible for coordinating with the Literary Circuit and organizing open mics, book launches, and literary competitions.

Core Member – Enactus DTU *Feb 2020 – Feb 2021*

- Led a team of 30+ members to implement an IoT and Tech Research-enabled Aquaponic system.
- Part of the team representing Milan as a startup in the IC^2 Business Plan Competition (finished 5th nationally)

Technical Skills

- **Programming Languages** : Python | **C/C++** | JavaScript | **Java** | Dart | Lua
- **Frontend** : HTML | CSS | JS | React.js | Flutter
- **Backend** : **SpringBoot** | Django | **Play** | MySQL | PostgreSQL
- **CS Fundamentals** : OOPS | **Data Structures and Algorithms** | OS | Computer Networks | **DBMS**